DATA SCIENCE & ANALYTICS

MODULE 03 SESSION 16

Review, four models and MLOps, then Power BI

A model people rely on is a process.

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Thu 29 Oct 2026

BY THE END OF TODAY, YOU CAN

Class objectives

01 Pick the model; score it honestly

02 Run five steps for four kinds of model

03 Say what MLOps is and how BigQuery does it

04 Say what your Power BI licence allows

05 Load, clean and chart a CSV file

INSTRUCTOR DO

Step 1.1

Two parts today

Machine learning first: a review, four models, MLOps, Assignment 2. Then Power BI.

ML · 01

Review of D11–D15

The ideas every model uses, on today’s new datasets. The earlier days hold the depth.

INSTRUCTOR DO

Step 2.1

Features, label and the four kinds of model

Features in, a label out. What the label looks like decides the model.

DEFINITION · D11

A model is a rule learned from past rows: it reads the features of one row and returns the label.

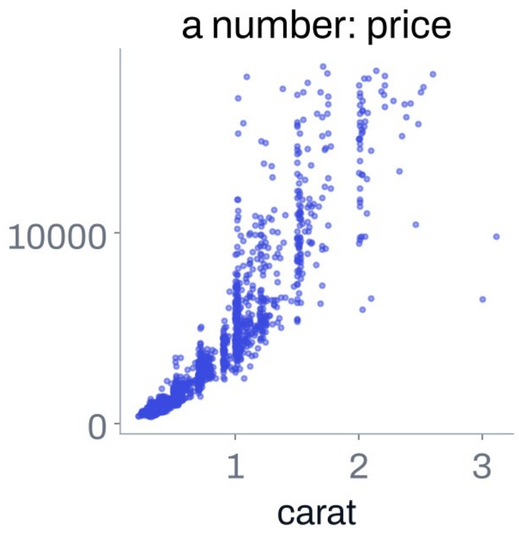
ONE ROW IS ONE PASSENGER · ml.titanic · TODAY'S DATA

Features and label in ml.titanic

passenger_id	pclass	sex	age	sibsp	parch	survived
1	1	female	29.0	0	0	**true**
2	1	male	0.92	1	2	**true**
3	1	female	2.0	1	2	**false**
501	2	male	46.0	0	0	**false**
502	2	female	13.0	0	1	**true**

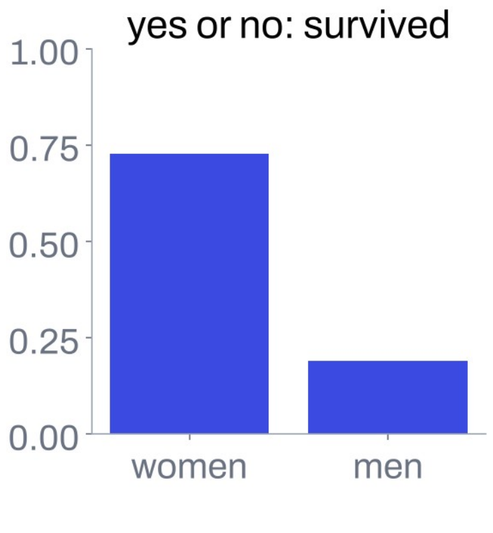
Features: pclass to parch. Label: survived (lime). passenger_id names the row; it is not a feature, so it is left out with EXCEPT.

The four kinds of model



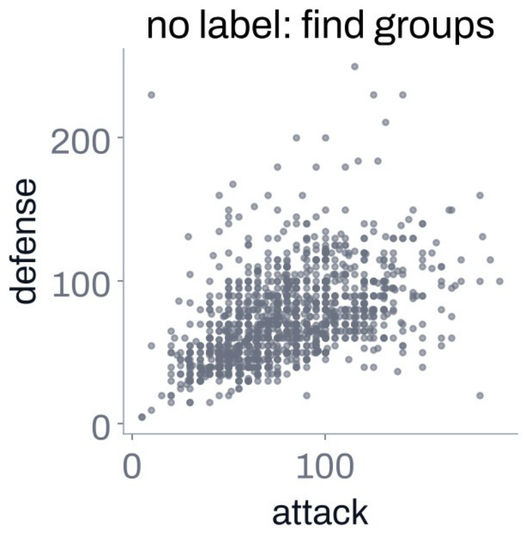
What is this diamond worth?

`'LINEAR_REG'`



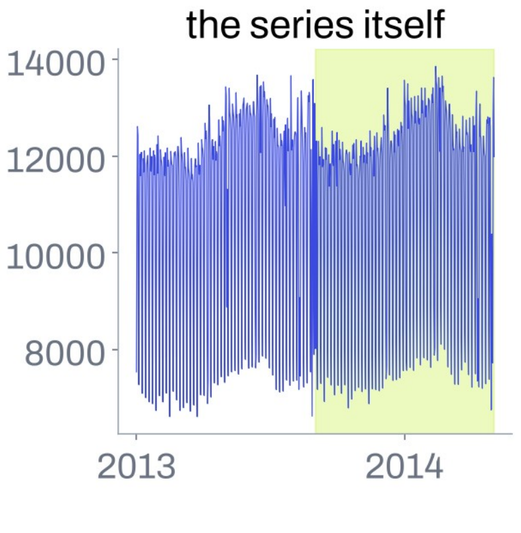
Who was likely to survive the Titanic?

`'LOGISTIC_REG'`



What kinds of Pokémon are there?

`'KMEANS'`



How many babies each day next year?

`'ARIMA_PLUS'`

A number, yes or no, no label, a series in time: four kinds of model, one routine.

INSTRUCTOR DO

Step 2.2

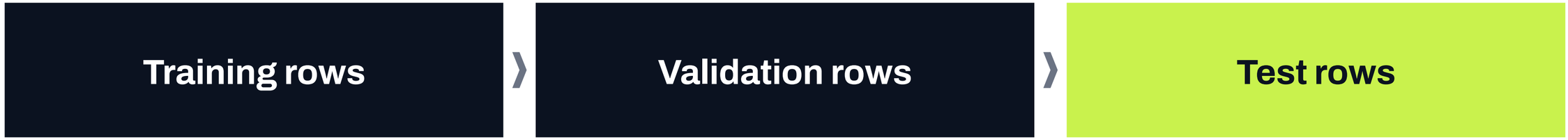
Evaluating a model

Score on rows the model never saw, and beat a simple guess first.

THE ROWS YOU SCORE ON · D11, D14, D15

Training, validation and test rows

Rows of examples: diamonds, Titanic



A series: births



On its own training rows, a model has seen the answers.

Each model against a simple guess

Kind	Score	Simple guess	Model
Diamonds	MAE: the typical miss, dollars	the median price: 2,808	**about 740**
Titanic	accuracy: share right	“nobody survived”: 0.62	**about 0.78**
Births	MAE: the typical miss, babies a day	same weekday a year before: 316	from BigQuery
Pokémon	Davies–Bouldin: lower is tighter	no right answer: compare k	about 1.6 at k = 4

Regression: MAE and R^2 (D11). Classification: accuracy against the majority guess (D12). Clustering: no label, so no accuracy (D13).

Precision, recall and threshold

Term	The question it answers
Precision	of the passengers flagged “survived”, how many did?
Recall	of the passengers who survived, how many were flagged?
Threshold	the probability above which we say TRUE: 0.5 unless you move it

D12: with 237 leavers among 1,470 employees, “nobody leaves” is 84% accurate and finds none of them. Lower the threshold to find more, at the cost of more false alarms.

INSTRUCTOR DO

Step 2.3

Trees, overfitting and explaining a prediction

When a straight line is not enough, trees. When a number surprises you, ask the model why.

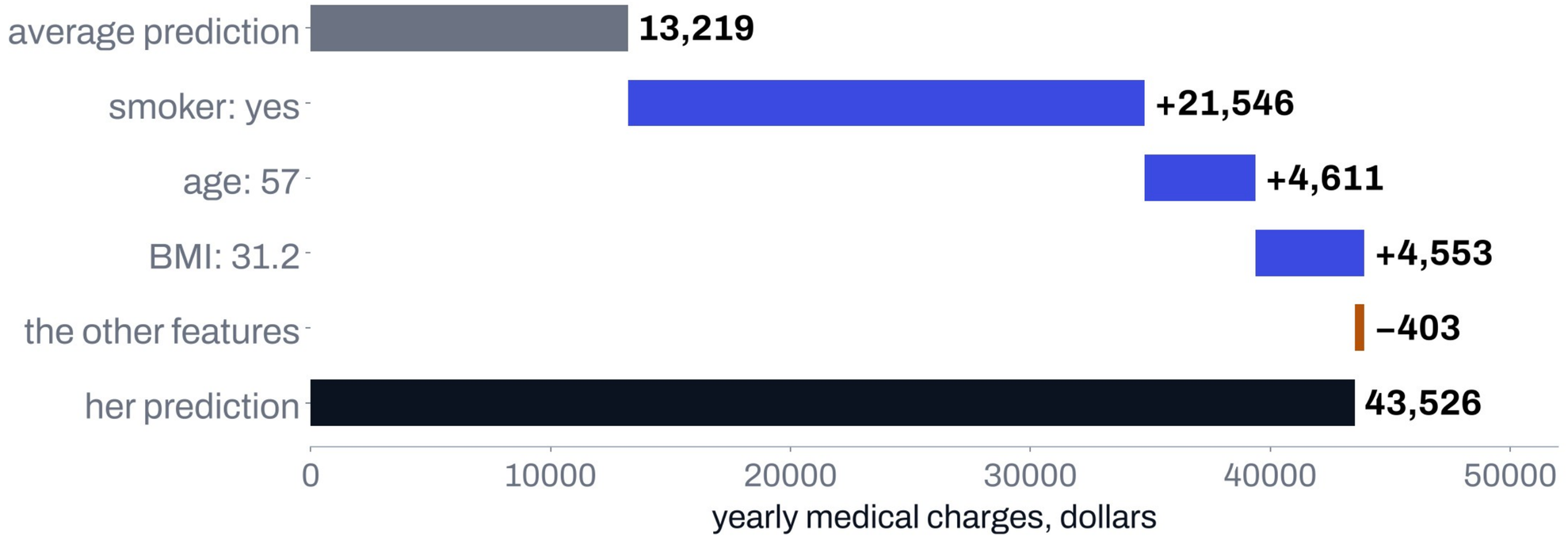
From a line to trees, and overfitting



Overfitting

The training score keeps rising while the score on unseen rows stops or falls: the model has memorised rows, not learned a rule. Choose on validation rows.

How the model reached 43,526 for one person



Start from the average. Each of her values adds or subtracts its contribution (not a weight). Together: 43,526. It shows what the model leaned on, not causes.

DEFINITION · D15

TRANSFORM saves the preparation rules with the model: one statement prepares the data and trains, and every later prediction applies the same rules.

EVERYONE DO

Step 2.4

Name the model

Four questions from four businesses. Which kind of model does each need?

QUICK CHECK · ANSWER IN SLACK

Which kind of model?

#	The question
1	How many calls will the help desk get each day next month?
2	Will this customer cancel the subscription?
3	What rent should this flat ask?
4	Which shoppers behave alike?

QUICK CHECK · THE ANSWER

Answers

#	The label	Kind	model_type
1	a number over time	forecast	**'ARIMA_PLUS'**
2	yes or no	classification	**'LOGISTIC_REG'**
3	a number	regression	**'LINEAR_REG'**
4	no label	clustering	**'KMEANS'**

WHERE TO GO DEEPER

Where each idea was taught

Day	The ideas	Redo
D11	features and label · regression · MAE and R^2	labs/D11
D12	classification · threshold · precision and recall	labs/D12
D13	clustering · choosing k · scaling	labs/D13
D14	trees · fixed groups · the test once	labs/D14
D15	forecasting · explaining a prediction · TRANSFORM	labs/D15

Today uses four new datasets, uploaded fresh, so nothing depends on an earlier table.

ML · 02

Four models in BigQuery ML

Four new datasets, so you can see the steps do not depend on the data: look, create, evaluate, predict, save.

INSTRUCTOR DO

Step 3.1

Five steps for every model

The four kinds from the review, one routine. Watch what changes: a word or two in OPTIONS.

EVERY MODEL, EVERY TIME

Five steps

Step	What you do	SQL
1 Look	one row is one ___; which column is the label?	SELECT … FROM ml.<table>
2 Create	train the model	CREATE OR REPLACE MODEL
3 Evaluate	score it on rows it did not learn from	ML.EVALUATE
4 Predict	one row in, a prediction out	ML.PREDICT · ML.FORECAST
5 Save	a view Power BI can read	CREATE OR REPLACE VIEW

BEFORE YOU START · ACTIVITY 01

Upload diamonds.csv into ml

shared/datasets/ml_recap on GitHub: one CSV per activity, into your own ml.

```
CREATE SCHEMA IF NOT EXISTS ml OPTIONS (location = 'US');
-- ml → ⋮ → Create table → Upload: diamonds.csv
-- File format CSV · Table diamonds · Auto detect
SELECT COUNT(*) FROM ml.diamonds;
```

→ 53,940 rows

EVERYONE DO

Step 3.2

What is this diamond worth?

53,940 diamonds: carat, cut, colour, clarity, size and the price they sold for. Regression.

labs/D16/01-Evr-Diamond-Prices/README.md

STEP 2 · CREATE

Train the diamond price model

`model_type = 'LINEAR_REG'` and **`EXCEPT (diamond_id)`**: the rest is the same as D11.

```
CREATE OR REPLACE MODEL ml.diamond_price
OPTIONS (model_type = 'LINEAR_REG',
         input_label_cols = ['price']) AS
SELECT * EXCEPT (diamond_id)
FROM ml.diamonds;
```

WHAT COMES BACK · STEPS 1 AND 3

Off by about 740 dollars on unseen diamonds

Check	Value
Diamonds · cheapest · dearest	53,940 · 326 · 18,823 dollars
Median price	about 2,401
Guess the median for every diamond: MAE	2,808
The model: MAE · R^2	**about 740 · about 0.92**

Unseen: BigQuery held some diamonds aside itself (the default split) and scored those. The model's typical miss is about a quarter of the guess's.

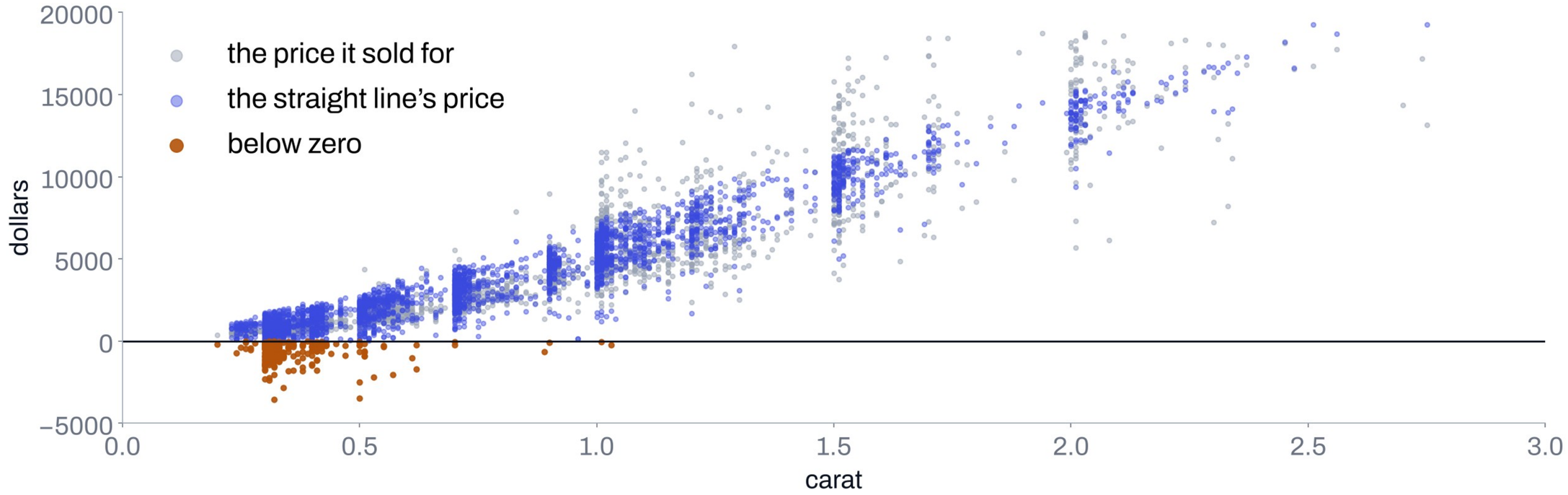
Four of the first five are priced below zero

diamond_id	carat	cut	price	predicted_price
1	0.23	Ideal	326	**about -1,346**
2	0.21	Premium	326	**about -665**
3	0.23	Good	327	about 211
4	0.29	Premium	334	**about -831**
5	0.31	Good	335	**about -3,459**

Small stones that sold for about 330 dollars. The next slide shows why the line goes below zero.

WHY · 3,000 OF THE 53,940 DIAMONDS

A straight line prices the smallest diamonds below zero



Real prices climb faster than a line. About 7 in 100 predictions fall below zero, 9 in 10 under half a carat; lab 06 fits the log of the price.

STEP 5 · SAVE

The predictions, as a view

CREATE OR REPLACE VIEW over ML.PREDICT: Power BI reads it by name on D22.

```
CREATE OR REPLACE VIEW ml.v_diamond_prices AS
SELECT diamond_id, carat, cut, color, clarity,
       price, ROUND(predicted_price) AS predicted_price
FROM ML.PREDICT(MODEL ml.diamond_price,
                TABLE ml.diamonds);
```

→ 53,940 diamonds · about 3,900 below zero

EVERYONE DO

Step 3.3

Who was likely to survive?

Upload titanic.csv first. 1,309 passengers; survived is TRUE or FALSE.

labs/D16/02-Evr-Titanic-Survival/README.md

STEP 2 · CREATE

Train the survival model

`'LOGISTIC_REG'` and `['survived']`: 263 empty ages are filled with the average.

```
CREATE OR REPLACE MODEL ml.titanic_survival
OPTIONS (model_type = 'LOGISTIC_REG',
         input_label_cols = ['survived']) AS
SELECT * EXCEPT (passenger_id)
FROM ml.titanic;
```

WHAT COMES BACK · STEPS 1, 3 AND 4

Better than “nobody survived”

Check	Value
Survived: women · men	339 of 466 (0.727) · 161 of 843 (0.191)
“Nobody survived” is right	62% of the time
The model: accuracy · ROC AUC	**about 0.78 · about 0.84**
17, female, first class, with a parent	about 0.95
20, male, third class, alone	about 0.14

The view ml.v_titanic_scored keeps the chance of TRUE for all 1,309 passengers.

STEP 5 · SAVE

Save the predictions as a view

WHERE p.label = TRUE: one probability per passenger, the chance of TRUE.

```
CREATE OR REPLACE VIEW ml.v_titanic_scored AS
SELECT passenger_id, pclass, sex, age, survived,
       predicted_survived,
       ROUND(p.prob, 3) AS survival_chance
FROM ML.PREDICT(MODEL ml.titanic_survival, TABLE ml.titanic),
     UNNEST(predicted_survived_probs) AS p
WHERE p.label = TRUE;
```

→ 1,309 rows · first-class women about 0.88

EVERYONE DO

Step 3.4

TRANSFORM and the evaluation charts

Two models on the same passengers, one with TRANSFORM. Then read and compare their charts.

labs/D16/02-Evr-Titanic-Survival/README.md

TRANSFORM prepares the data, then trains, in one statement

Training



Prediction



Without TRANSFORM, every query using the model repeats the preparation.

BEFORE · FIVE TRAINING PASSENGERS FROM ml.titanic

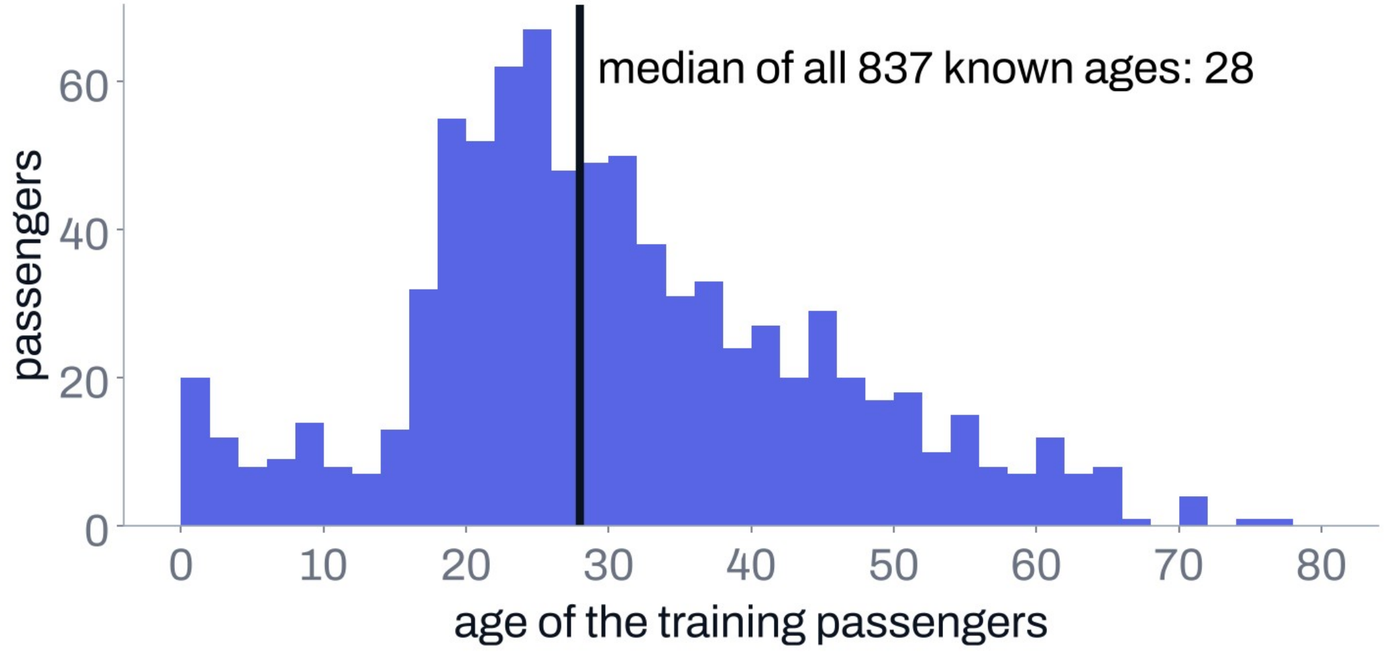
The raw rows have empty ages and no family size

passenger	pclass	sex	age	sibsp	parch
16	1	male	**empty**	0	0
682	3	female	**empty**	0	2
274	1	male	6	0	2
7	1	female	63	1	0
34	1	female	58	0	0

Three rules to apply: fill an empty age; mark children under 13; count the family.

RULE 1 · ML.IMPUTER(age, 'median') OVER ()

An empty age becomes 28, the median of the training ages



ML.IMPUTER(age, 'median')

fill empty ages with the median

OVER ()

look across all 1,048 training rows

Saved

28 is kept in the model and used again at prediction

Half the passengers with a known age are younger than 28, half older.

AFTER · THE SAME FIVE PASSENGERS

What the model trains on

passenger	pclass	sex	age	is_child	family_size
16	1	male	**28**	**FALSE**	**1**
682	3	female	**28**	**FALSE**	**3**
274	1	male	6	**TRUE**	**3**
7	1	female	63	**FALSE**	**2**
34	1	female	58	**FALSE**	**1**

is_child: IF(age < 13, TRUE, FALSE). family_size: sibsp + parch + 1, the passenger included. Lit: what TRANSFORM made.

One statement: the rules, then the model

TRANSFORM: the three rules from the last three pages.

```
CREATE OR REPLACE MODEL ml.titanic_transform
TRANSFORM (pclass, sex,
  ML.IMPUTER(age, 'median') OVER () AS age,
  IF(age < 13, TRUE, FALSE) AS is_child,
  sibsp + parch + 1 AS family_size,
  survived)
OPTIONS (model_type = 'LOGISTIC_REG',
         input_label_cols = ['survived'],
         data_split_method = 'NO_SPLIT') AS
SELECT pclass, sex, age, sibsp, parch, survived
FROM ml.titanic WHERE MOD(passenger_id, 5) <> 0;
```

The model applies its saved rules to a raw row

You send

pclass 3 › male › age empty › sibsp 1 › parch 3

Saved rules

age 28	is_child FALSE	family_size 5
training median	not under 13	1 + 3 + 1

Weights

chance 0.04 › did not survive

STEP 10 · THE SAME 261 PASSENGERS · 97 SURVIVED

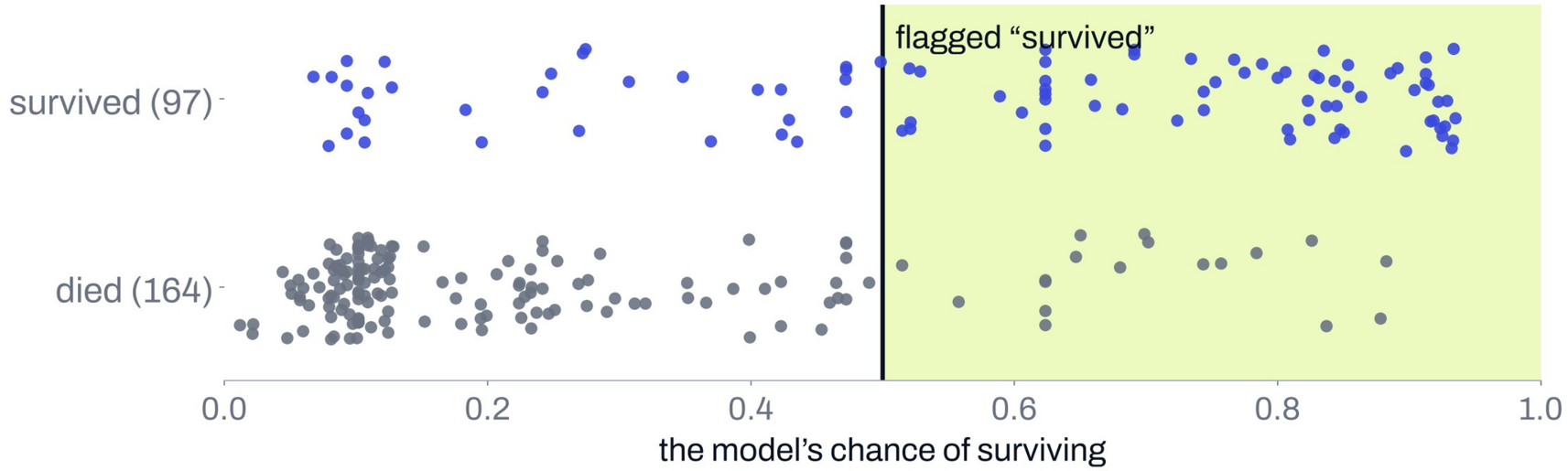
Sex only against the TRANSFORM model

Model	Accuracy	Precision	Recall	ROC AUC
sex only	about 0.78	about 0.73	about 0.63	about 0.75
TRANSFORM	about 0.81	about 0.78	about 0.67	**about 0.85**

Accuracy, precision and recall are counted at one threshold, 0.5. The next pages count them at every threshold.

ONE THRESHOLD, COUNTED · TRANSFORM MODEL · 261 PASSENGERS

At 0.5 the model flags 83 passengers: 65 survived, 18 died



Precision = 65 ÷ 83 = 0.78

of those flagged, the share who survived

Recall = 65 ÷ 97 = 0.67

of all 97 survivors, the share flagged

FPR = 18 ÷ 164 = 0.11

false positive rate: of all 164 who died, the share flagged

THE FOUR COUNTS · THRESHOLD 0.5 · 261 PASSENGERS

Each passenger lands in one of four boxes

	Flagged “survived”	Not flagged
Survived (97)	**65 · true positives**	**32 · false negatives (missed)**
Died (164)	**18 · false positives (false alarms)**	**146 · true negatives**

Lime: the model was right. Amber: the model was wrong.

PRECISION · AGAINST EVERYONE FLAGGED

Precision: how often a “survived” flag is right

Of the 83 passengers we flagged, how many survived?



$$\text{Precision} = \frac{\text{65 true positives}}{\text{65 + 18, everyone flagged}} = 0.78$$

Read it: when the model says “survived”, it is right 78 times in 100. A false alarm lowers it.

RECALL · AGAINST EVERYONE WHO SURVIVED

Recall: how many survivors the model finds

Of the 97 passengers who survived, how many did we flag?



$$\text{Recall} = \frac{\text{65 true positives}}{\text{65 + 32, every survivor}} = 0.67$$

Read it: the model finds 67 of every 100 survivors. A missed survivor lowers it. Also called the true positive rate.

FALSE POSITIVE RATE · AGAINST EVERYONE WHO DIED

False positive rate: who died but was flagged

Of the 164 passengers who died, how many did we flag?



$$\text{FPR} = \frac{\text{18 false positives}}{\text{18 + 146, everyone who died}} = 0.11$$

Read it: 11 in 100 who died were flagged by mistake. Lower is better. The ROC curve puts it across.

THREE THRESHOLDS, COUNTED THE SAME WAY

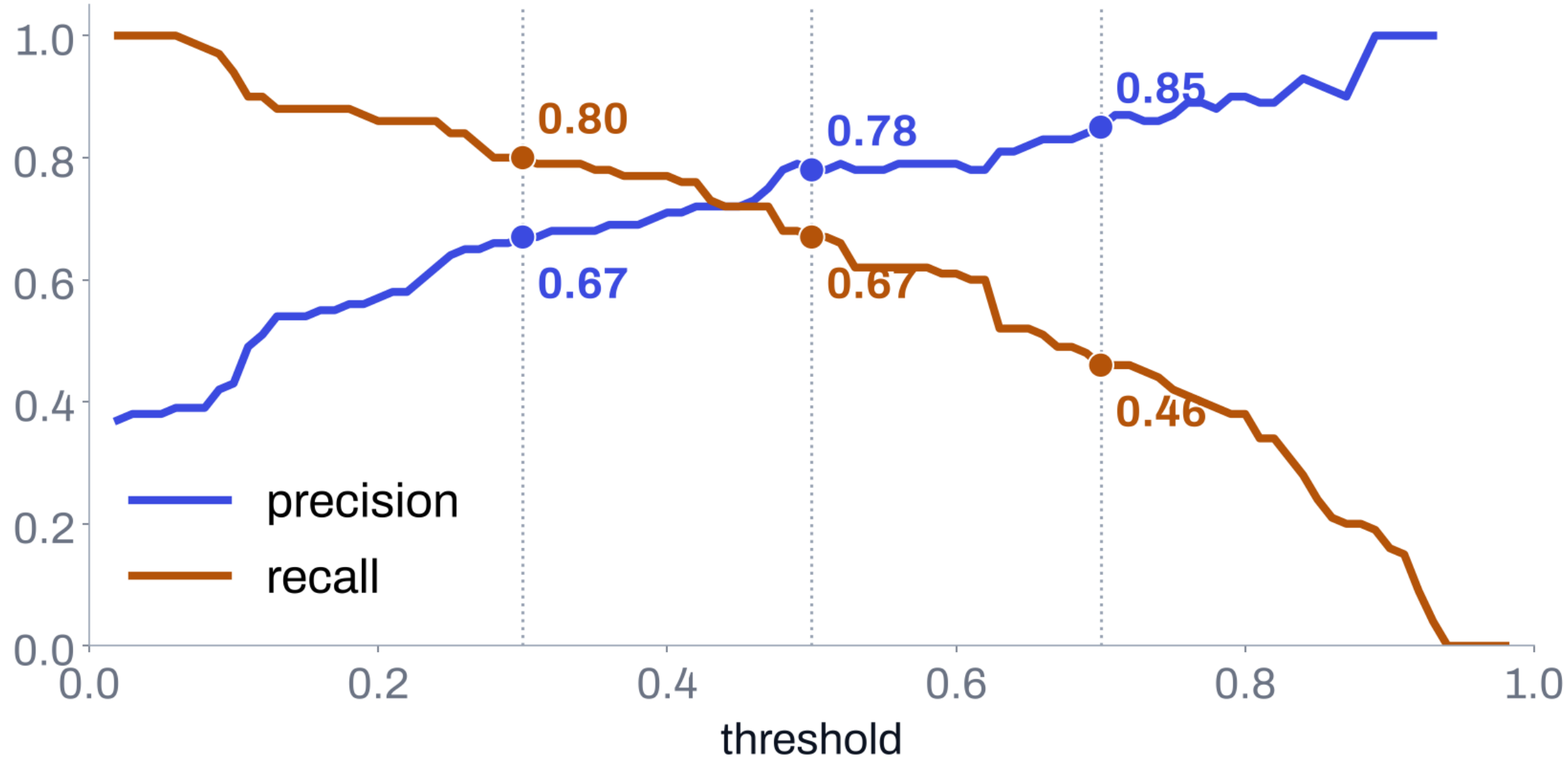
A lower threshold finds more survivors and more false alarms

Threshold	Flagged	Survivors found	False alarms	Precision	Recall	False pos. rate
0.3	116	78	38	0.67	0.80	0.23
0.5	83	65	18	**0.78**	**0.67**	**0.11**
0.7	53	45	8	0.85	0.46	0.05

Precision and recall pull against each other: no threshold gives the best of both. The business decides which mistake costs more.

CHART 1 · THE TABLE, AT EVERY THRESHOLD

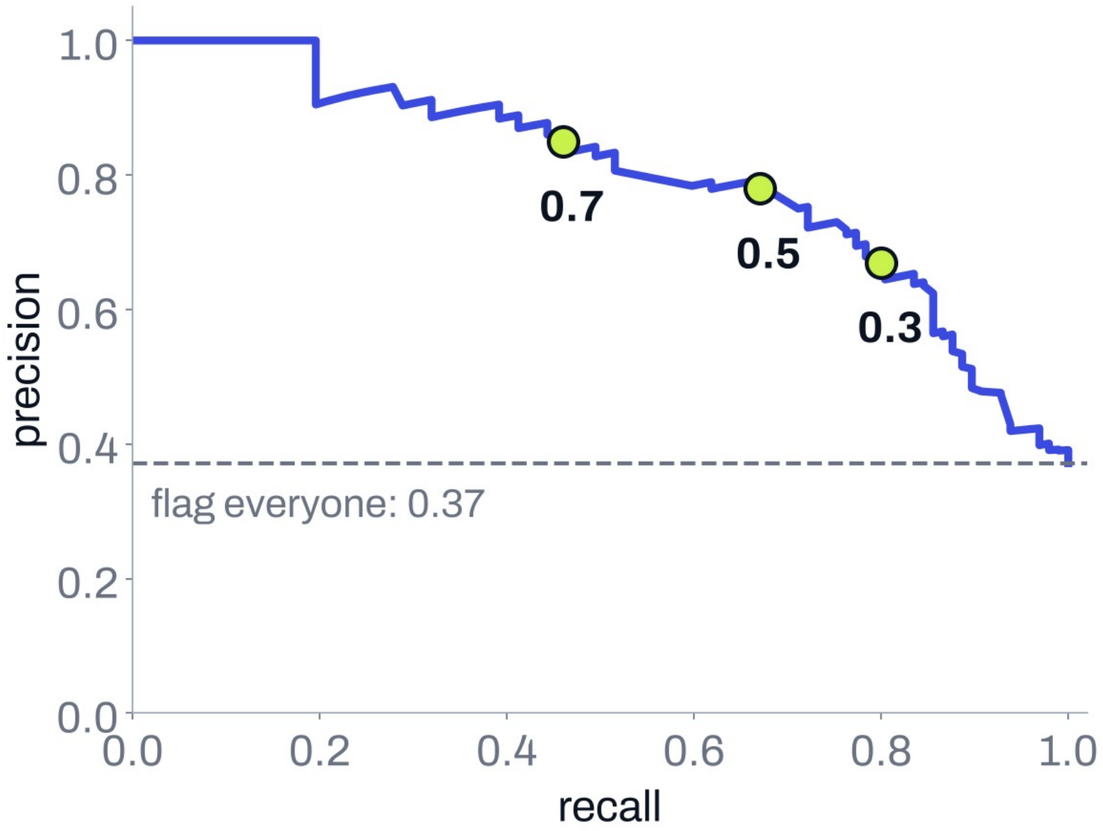
Precision and recall by threshold



Move right: precision rises, recall falls. Choose the threshold where the trade suits the decision.

CHART 2 · PRECISION-RECALL CURVE

Precision-recall curve: one point per threshold



The three dots

the table’s thresholds: 0.3, 0.5, 0.7

Good

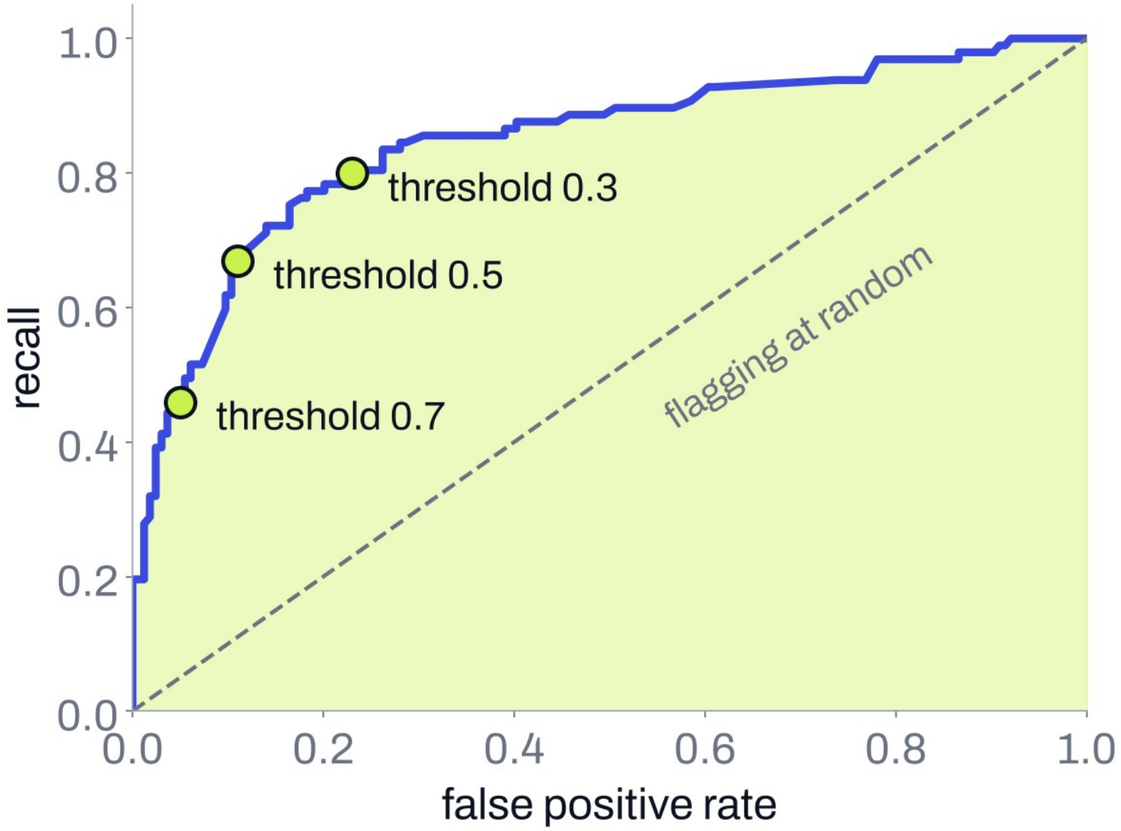
high precision while recall grows: towards the top right

Dashed line

flag everyone: precision is the share who survived, 0.37

CHART 3 · ROC CURVE

ROC curve: one point per threshold



The three dots

the same thresholds: at 0.5, 0.11 across and 0.67 up

Good

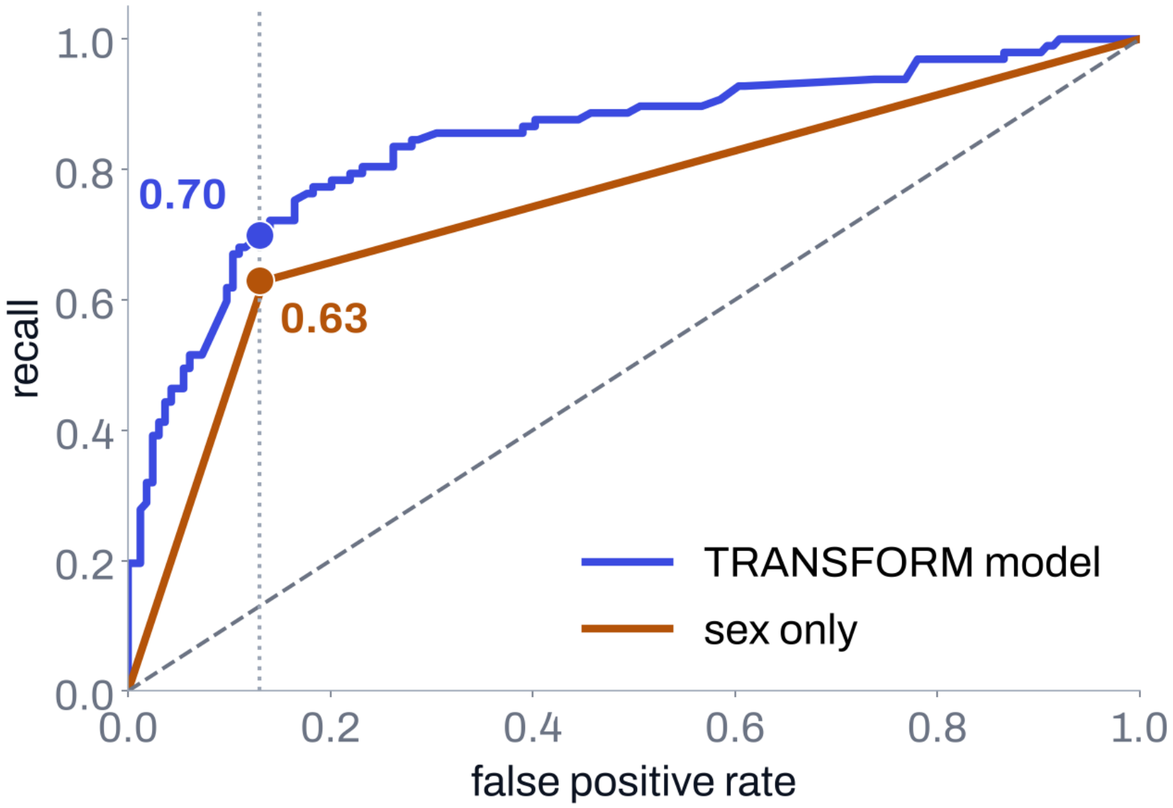
many survivors found for few false alarms: towards the top left

Area, ROC AUC 0.85

pick one survivor and one who died: 85 times in 100 the survivor gets the higher chance

COMPARING TWO MODELS · ROC · THE SAME 261 PASSENGERS

Same false alarms: TRANSFORM finds more survivors



Sex only: flag every woman

0.13 of those who died flagged, 0.63 of survivors found

TRANSFORM, same false alarms

0.70 of survivors found

The rule

score both on the same rows; the curve that is higher everywhere wins

EVERYONE DO

Step 3.5

What kinds of Pokémon are there?

Upload pokemon.csv first. 1,215 Pokémon, six stats, no label.

labs/D16/03-Evr-Pokemon-Groups/README.md

STEP 2 · CREATE

Train the k-means model

'KMEANS' and **num_clusters = 4**: no input_label_cols at all.

```
CREATE OR REPLACE MODEL ml.pokemon_groups
OPTIONS (model_type = 'KMEANS',
         num_clusters = 4) AS
SELECT hp, attack, defense,
       sp_attack, sp_defense, speed
FROM ml.pokemon;
```

Four kinds of Pokémon

Pokémon	What stands out	Total	A name	For example
about 396	every stat about 50	about 300	**first-stage**	Pikachu, Magikarp
about 332	Speed 96, Attack 92	about 478	**fast attackers**	Gengar
about 263	Defense 105, Speed 50	about 483	**tanks**	Snorlax
about 224	Special Attack 118	about 597	**powerhouses**	Charizard, Mewtwo

BigQuery numbers the groups at random: the view keeps centroid_id, and the names go in the report.

EVERYONE DO

Step 3.6

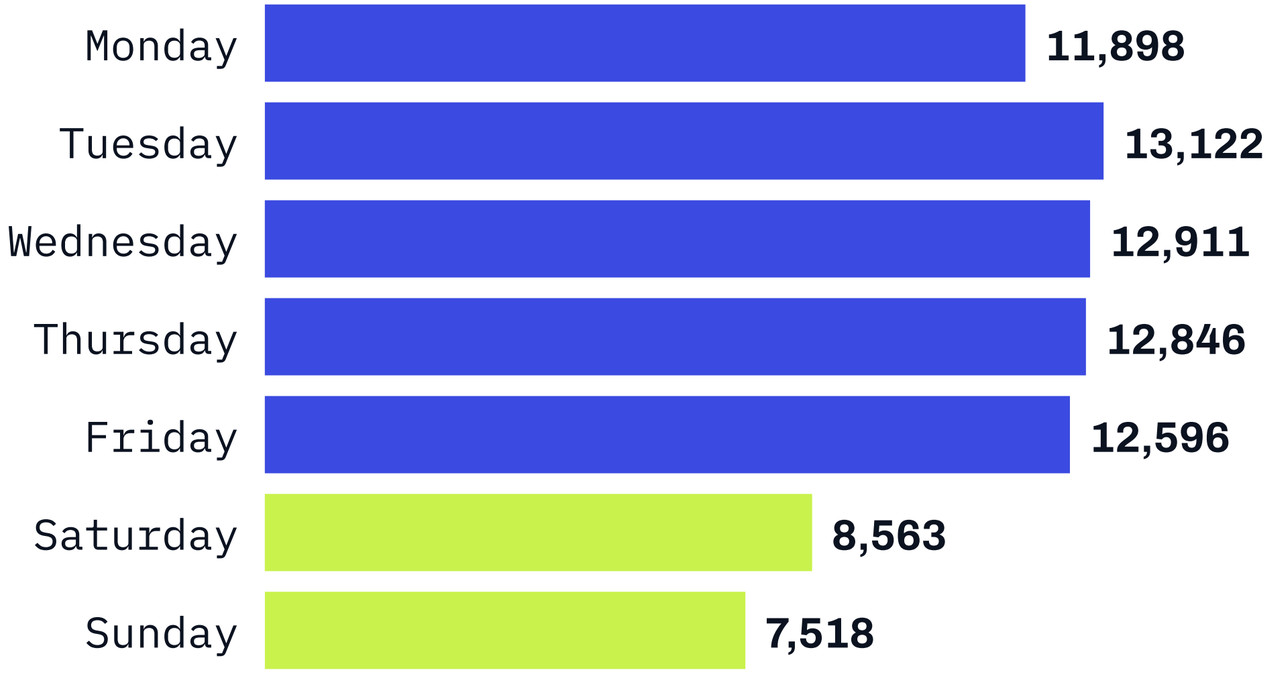
How many babies each day?

Upload us_births.csv first. Train on 2000–2013, forecast 2014, append the forecast to the history.

labs/D16/04-Evr-Births-Forecast/README.md

WHAT COMES BACK · STEP 1 · AVERAGE BIRTHS A DAY · ALL 5,479 DAYS, 2000–2014

Births follow the working week



FRIDAYS ONLY

Friday the 13th (25 of them)

11,950

other Fridays

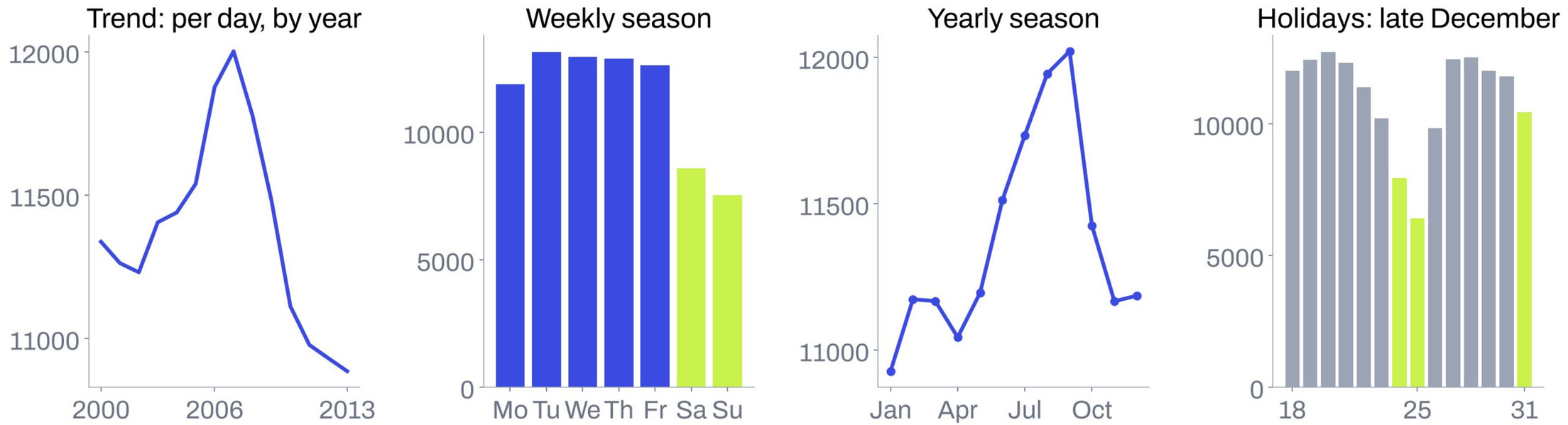
12,618

Planned births avoid weekends and, it seems, the 13th.

So a forecast must carry the weekly rhythm: one number for every day would be thousands off each Sunday.

ARIMA_PLUS · DAILY US BIRTHS, 2000–2013 · AVERAGES FROM THE DATA

ARIMA_PLUS learns a trend, seasons and holidays



It learns each piece from 2000–2013 and carries it into 2014. Rows have an order, so it is scored on the later year, not a random sample.

STEP 2 · CREATE

Train the forecast model

`holiday_region = 'US'`: 4 July and Christmas get their own effect.

```
CREATE OR REPLACE MODEL ml.births_arima
OPTIONS (model_type = 'ARIMA_PLUS',
  time_series_timestamp_col = 'birth_date',
  time_series_data_col = 'births',
  horizon = 365, holiday_region = 'US') AS
SELECT birth_date, births FROM ml.us_births
WHERE birth_date < '2014-01-01';
```

Step 3 scores 2014. The bar: “same weekday a year earlier” misses by 316 a day.

STEP 5 · SAVE · THE FORECAST APPENDED TO THE HISTORY

History and forecast in one view

UNION ALL stacks them; series says which half a row comes from.

```
CREATE OR REPLACE VIEW ml.v_births_history_forecast AS
SELECT birth_date AS day, 'actual' AS series,
       CAST(births AS FLOAT64) AS births, ...
FROM ml.us_births
UNION ALL
SELECT DATE(forecast_timestamp), 'forecast',
       ROUND(forecast_value), ...
FROM ML.FORECAST(MODEL ml.births_arima,
       STRUCT(365 AS horizon, 0.95 AS confidence_level));
```

→ actual 5,479 days · forecast 365 days

ML · 03

MLOps

What MLOps is, why a saved model is not enough, and how BigQuery ML covers each step.

INSTRUCTOR DO

Step 4.1

What MLOps is

The pricing team opens their report next month. Which model? Trained when? Still right?

THE PRICING TEAM, NEXT MONTH

Three questions about ml.v_diamond_prices

Question	Where the answer has to come from
Which model made these numbers?	a model name that does not change under the report
When was it trained, and on which rows?	a record kept with the model
Are the numbers still right?	a check of new data against the training data

The habits and tools that answer these three questions are what people call MLOps.

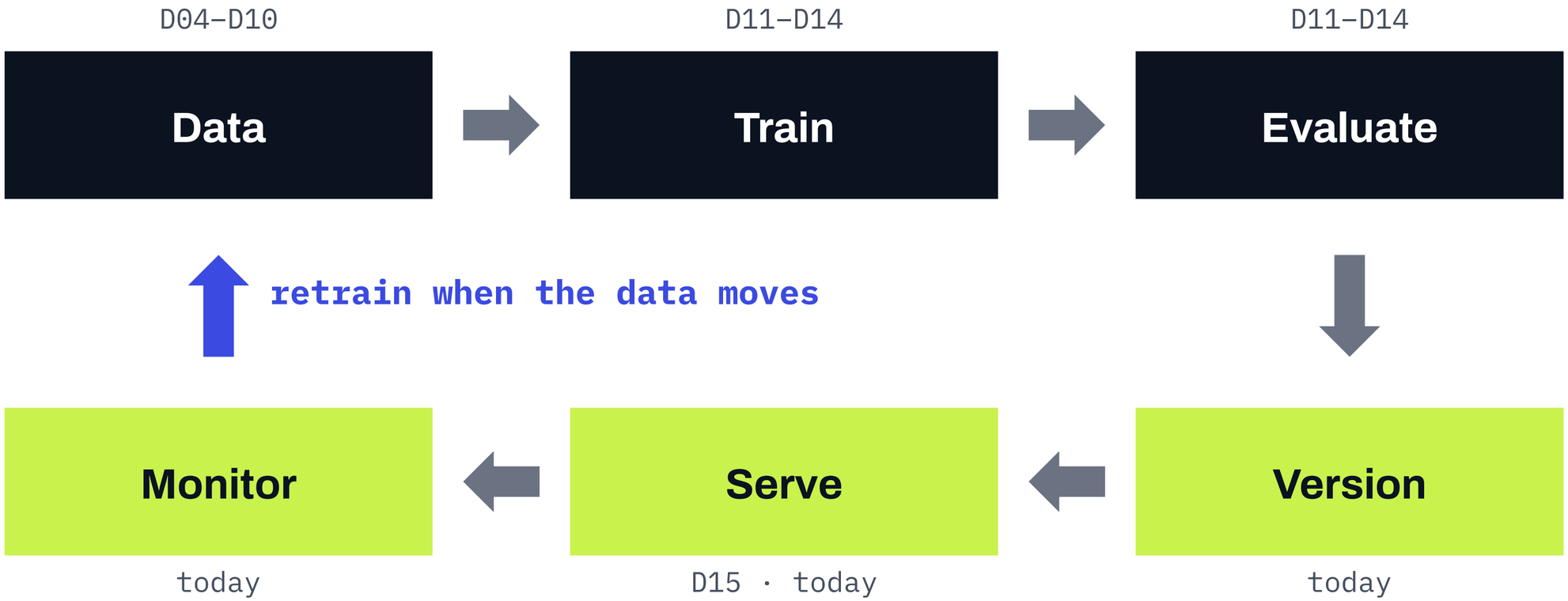
MLOps is the set of practices that keep a model working after the first query: repeatable training, versioned models, predictions served where people use them, monitoring, and retraining.

Five things that go wrong without it

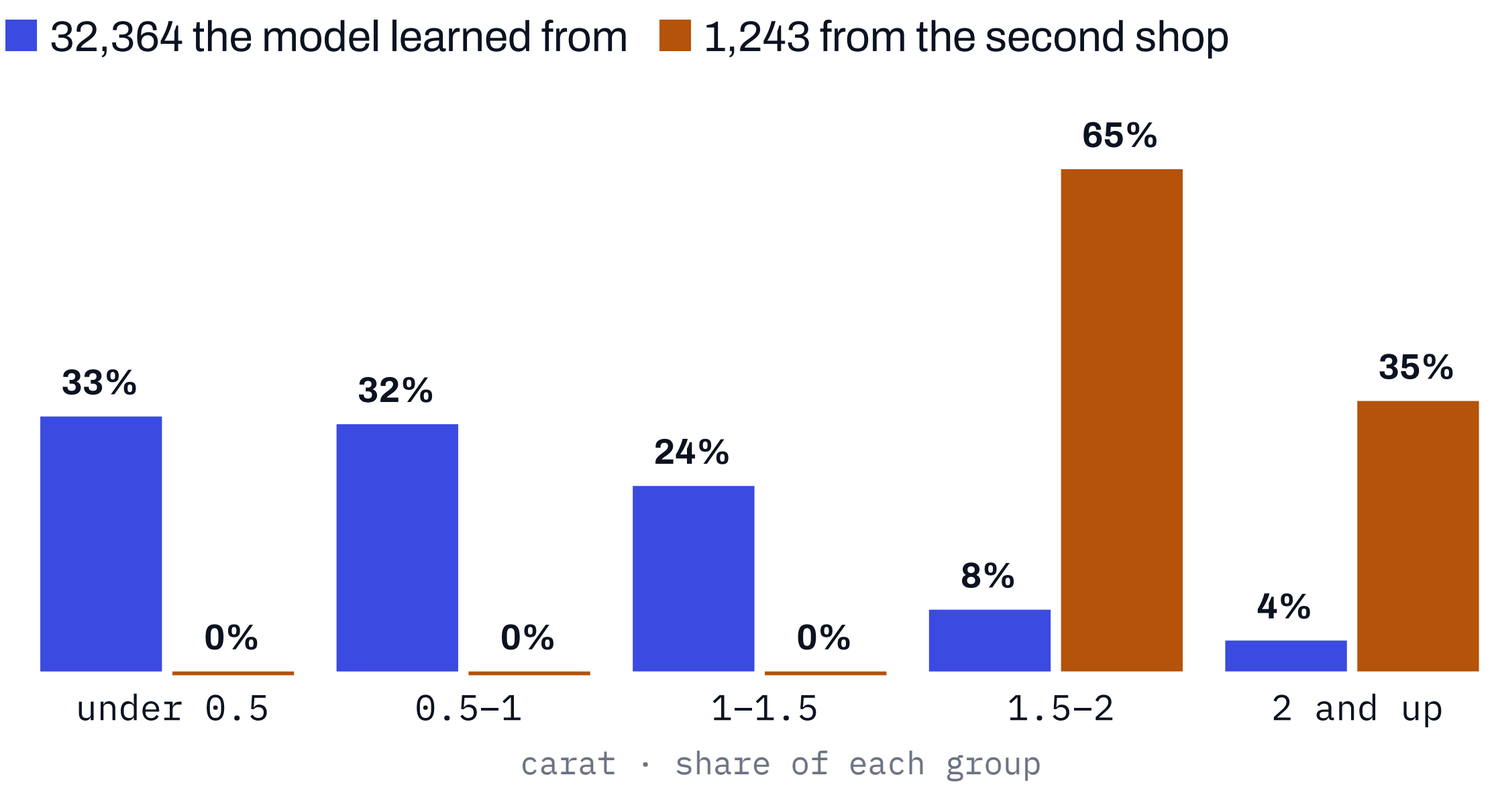
What goes wrong	In this course
The world moves, the model does not	your diamond model meets bigger stones
Nobody can say which version made the numbers	CREATE OR REPLACE MODEL overwrites the old one
Training and serving apply different rules	TRANSFORM keeps them in the model (review)
Every retrain breaks the report	Power BI reads a table or view by its name
One model is a query; fifty are a system	a bank runs many, retrained every month

THE LIFECYCLE · A LOOP, NOT A LINE

The model lifecycle



New stones the model rarely saw: it misses twice as far



ML.VALIDATE_DATA_SKEW

flags carat, length, width and height before a single price is known

ML.EVALUATE, LATER

typical miss, once the prices are known

ordinary stones

about 742

big stones

about 1,764

Skew: new rows unlike the training rows. Drift: this month’s rows unlike last month’s. Nothing errors; the prices are quietly worse.

QUICK CHECK

A shop that sold small stones starts selling big ones

The price model learned from the small stones. Is that skew or drift?

Skew

new rows against the training rows

Drift

this month’s rows against last month’s

QUICK CHECK · THE ANSWER

Both: it depends on what you compare

Skew — the new stones against the training stones

→ **carat is flagged**

Drift — this month’s stones against last month’s

→ **carat moved**

Either way, the model prices stones it rarely saw.

INSTRUCTOR DO

Step 4.2

MLOps in BigQuery ML

One row per step of the loop: the BigQuery feature, and whether our sandbox can run it.

BigQuery ML features by lifecycle step

Step	BigQuery ML	Sandbox
Data	tables, views · ML.DESCRIBE_DATA	yes
Train	CREATE MODEL · TRANSFORM	yes
Evaluate	ML.EVALUATE on unseen rows · the test once	yes
Version	Details tab · ML.TRAINING_INFO · labels · a model registry	**no registry**
Serve	batch: ML.PREDICT to a table or view · online: an endpoint	**batch only**
Monitor	ML.VALIDATE_DATA_SKEW · ML.VALIDATE_DATA_DRIFT	billed as a query
Retrain	on a schedule: scheduled queries	**no**

Batch: predictions for many rows at once, into a table or view. Online: one prediction on request, from an endpoint. Scheduled queries need a service the sandbox does not have.

A monitoring script in SQL

ML.VALIDATE_DATA_SKEW compares new rows with the training statistics.

```
IF (SELECT COUNTIF(is_anomaly)
    FROM ML.VALIDATE_DATA_SKEW(MODEL ml.diamond_price,
         (SELECT * FROM new_batch))) > 0 THEN
  CREATE OR REPLACE MODEL ml.diamond_price ...;
ELSE
  SELECT * FROM ML.PREDICT(MODEL ml.diamond_price,
         (SELECT * FROM new_batch));
END IF;
```

ml.diamond_price: your model from 3.2. new_batch: this month’s stones.

What our sandbox does not do

Not supported	So in this course
DML: INSERT, UPDATE	save results with CREATE OR REPLACE TABLE … AS SELECT, or a view
BigQuery Data Transfer Service	no scheduled queries: we retrain and check by hand
Tables and views expire after 60 days	fine for a course that ends on 12 November

EVERYONE DO

Step 4.3

Model details, and table or view

Open the diamond model’s tabs, read ML.TRAINING_INFO, then set a table beside the view.

labs/D16/05-Evr-Model-Paper-Trail/README.md

Table or view

CREATE OR REPLACE TABLE: rows written once. The view runs ML.PREDICT at every read.

```
CREATE OR REPLACE TABLE ml.diamond_prices_snapshot AS
SELECT * FROM ml.v_diamond_prices;
```

→ today, the same average as the view

A retrain reaches the view, not the table

Diamond 1: 0.23 carat, Ideal, sold for 326 dollars

Monday · version 1 trained	Tuesday · version 2 retrained
a straight line on the price	a line on the log of the price: never below zero (lab 06)
a table, written Monday	
about −1,346 dollars	**about −1,346 dollars**
a view, read each day	
about −1,346 dollars	**about 330 dollars**

A table stores rows written once. A view stores the query, so ML.PREDICT runs at every read (Google: “the query that defines the view is run each time the view is queried”).

THE RULE TO KEEP

A model is finished when someone can trust its numbers next month.

A model scored once, for one decision, needs none of this. A model a report reads every week needs all of it.

INSTRUCTOR DO

Step 5.1

Assignment 2

The machine-learning module is complete. A2 is three models on one business, judged the way the business would judge them.

ASSIGNMENT 2 · MACHINE LEARNING · 20%

More than a third of our bookings never arrive

Part	What you hand in
Data	Hotel bookings: two hotels in Portugal, 119,390 bookings (CC BY 4.0)
Questions	who will cancel? (classification) · a fair price per night? (regression) · what kinds of guests? (clustering)
Hand in	your SQL, a screenshot per question, the write-up and a two-page memo to the revenue manager
Graded on	the reasoning: is a good score a good model? 40 of 100 points are the memo

Due before 4:00 PM, Thursday 5 November.

BI · 01

Power BI, Fabric and your licence

What Power BI is, where you work, and what your account lets you do.

INSTRUCTOR DO

Step 6.1

What Power BI is

Three parts, one path from data to a report, and the exam this block prepares for.

Desktop, the service, and mobile

Part	What it is for	Runs on
Power BI Desktop	building reports: get data, Power Query, model, visuals	Windows 10 or later
The Power BI service	publishing, sharing, refresh; building in the browser too	any browser: app.powerbi.com
Power BI Mobile	reading reports on a phone	iOS, Android

Desktop is free. There is no Mac version: Microsoft lists “Windows 10 or Windows Server 2016 or later”.

Other BI tools you will meet

Tool	A useful starting point	A Nakheel question
Tableau	Explore data visually	How does revenue compare by channel?
Qlik	Explore related and excluded data	Which stores have no completed sales?
Looker	Define business metrics centrally	Can every team use the same revenue definition?
ThoughtSpot	Ask data questions in plain language	“Completed revenue by channel?”

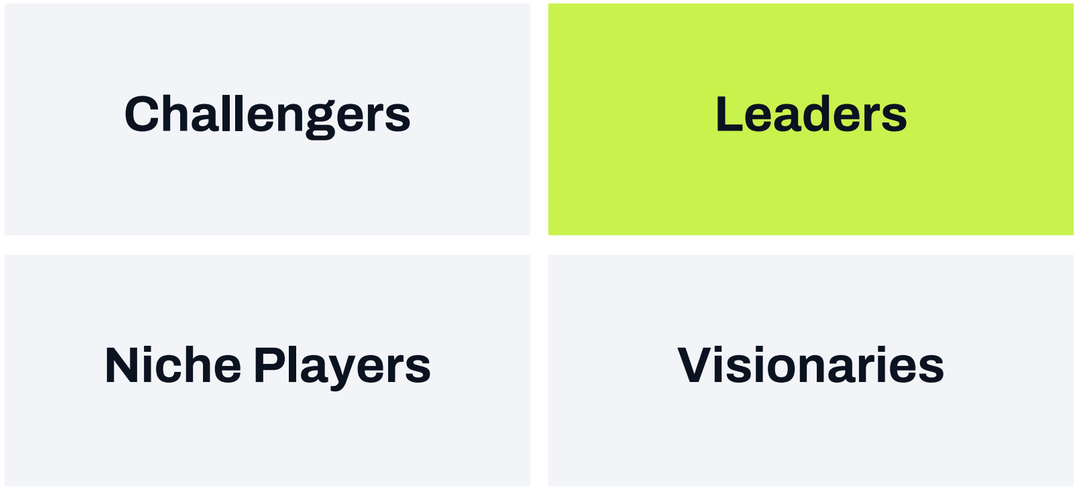
Capabilities overlap. The data and definitions still need checking.

GARTNER · ANALYTICS AND BUSINESS INTELLIGENCE · 2026

Microsoft: a Gartner Leader

READING GUIDE · NO VENDOR POSITIONS

Ability to Execute · delivery today ⬆



Completeness of Vision
market direction ➡

Choose for the company's needs.

Why companies choose Power BI

- Works with Microsoft 365
- Prepare, model and report
- Free Desktop; licensed sharing

Sources: Gartner 2026 report · Microsoft Power BI

THE PATH EVERY REPORT TAKES

From data to a report people use



Today: the lime boxes, once, quickly.

Power Query is the editor where you clean and shape data before it loads. Every click is saved as a named step, replayed at each refresh.

A semantic model is what a report reads: the tables, how they relate, and the measures such as revenue.

PL-300 · SKILLS MEASURED AS OF 20 APRIL 2026

The exam, in four parts

Part of the exam	Weight	Course days	Microsoft Learn path
Prepare the data	25–30%	D16–D17	Prepare data for analysis with Power BI
Model the data	25–30%	D18–D20	Model data with Power BI
Visualize and analyze the data	25–30%	D21–D22	Design effective reports in Power BI
Manage and secure Power BI	15–20%	D23	Manage and secure Power BI

INSTRUCTOR DO

Step 6.2

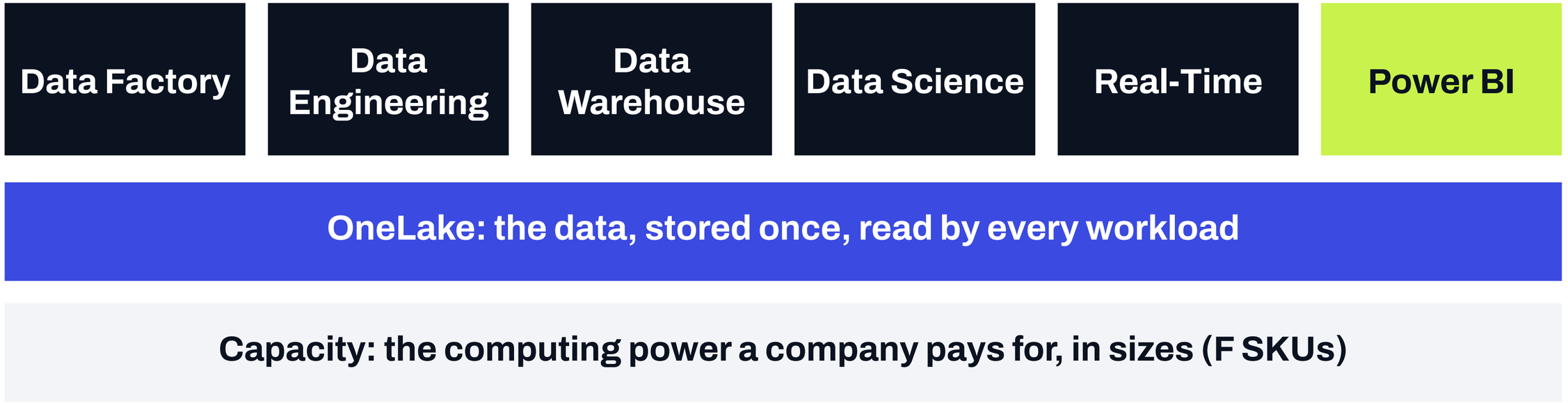
Fabric, and what each licence lets you do

What Microsoft Fabric is, what changed for Power BI, and what your university account allows.

DEFINITION

Microsoft Fabric is Microsoft’s analytics platform, sold as one online service: data engineering, warehousing, data science, real-time analytics and Power BI over one shared store of data, OneLake.

Power BI is one of Fabric's workloads



WHAT CHANGED FOR A POWER BI USER

Power BI now lives inside Fabric

Before	Now
the Power BI service	the same service, opened inside Fabric
Power BI Premium capacity (P SKUs)	Fabric capacity (F SKUs); “Microsoft no longer sells new P SKUs”
the free Power BI licence	Fabric (Free)
models edited only in Desktop	semantic models edited in the browser too

LICENCES · US LIST PRICES · CHECKED 1 NOVEMBER 2026

What each licence lets you do

	Build reports in My workspace	Publish to a shared workspace	Price
Fabric (Free)	✓	×	free
Power BI Pro	✓	✓	$14 / user / month
Premium Per User	✓	✓	$24 / user / month
Trial, if offered	✓	✓	check your account

Pro and PPU prices are paid yearly. Free covers personal work in My workspace; your university controls access.

TRIALS · CHECK YOUR ACCOUNT

Two kinds of trial

Trial	What it gives you	What to check
Power BI individual trial	paid Power BI features for your account	the features and end date shown
Fabric capacity trial	shared computing for Fabric workloads; Power BI trial rights may be included	60 days, if you are eligible

Your university can disable trials. You can use Desktop for today’s files without starting one.

EVERYONE DO

Step 6.3

Check your licence

Open app.powerbi.com and read your licence. If you choose an offered trial, check its end date.

EVERYONE DO · FOUR CLICKS

Your licence, and the trial

#	Do this	You should see
1	app.powerbi.com, university account	the Power BI home page
2	profile picture → Account manager	Free account, Pro, Trial, or Start trial
3	Trial offered? Read its terms before starting	the trial type and end date
4	post your access in Slack	Trial / Free / Pro; can you open Power Query?

Break

Stretch, then Power BI

After the break: from a CSV file to a chart that matches SQL.

BI · 02

From a CSV file to a chart

Load Nakheel’s customers and orders, clean one column in Power Query, and match the SQL revenue.

INSTRUCTOR DO

Step 7.1

Load, shape, show

Get data, Power Query, a bar chart and a card: the whole path once, quickly.

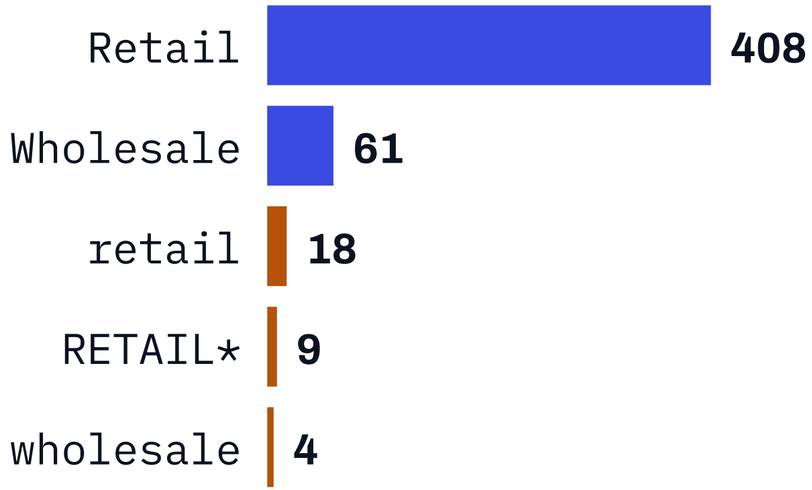
power-query/labs/D16/10-Evr-PowerBI-End-To-End/README.md

Every click becomes a step

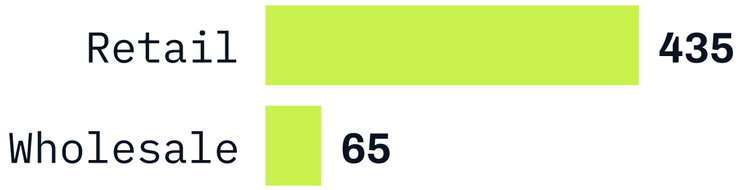
In Power BI	What it does	SQL you know
Get data → Text/CSV → Transform Data	opens Power Query on the file	the upload
View → profiling tools (browser: Data view)	errors, empties, distinct values	COUNT, GROUP BY
Transform → Format → Trim	removes spaces at the ends	TRIM(segment)
Transform → Format → Capitalize Each Word	retail → Retail	INITCAP(…)
Applied steps	the clicks, in order, replayed at refresh	a chain of CTEs

Two steps turn five spellings into two segments

POWER QUERY PROFILE · five raw spellings



AFTER · Trim + Capitalize · two spellings



The * marks a trailing space. 408 + 18 + 9 = 435; 61 + 4 = 65. The model groups case variants; check cleaning in the query profile.

WHAT COMES BACK · A CARD ON SUM OF order_total · ALL 3,000 ORDERS

The card adds every order; revenue is the completed part

No filter: **340,370.00**



cancelled 21,998.50 · pending 2,348.00

No filter	the card says 340,370.00 (2,771 + 212 + 17 orders)	×
Filters on this visual: completed	the card says 316,023.50 (2,771 orders)	✓
In SQL (D04)	WHERE status = 'completed': 316,023.50	✓

Show all digits: Callout → select Sum of order_total → Value → Display units: None. Other versions may say Callout values.

IN THE BROWSER · FIRST IMPORT

Start a report with customers.csv

Do this	What you should see
+ New Report → Get data → Text/CSV	the connection window
Upload file → customers.csv → Next	a preview; Next opens Power Query
Trim, then Capitalize Each Word on segment	Retail 435 · Wholesale 65
Create a report → My workspace	name the semantic model D16-nakheel

The report draws charts from D16-nakheel. Save the report as D16-nakheel-first-report.

IN THE BROWSER · ADD ANOTHER TABLE

Keep customers and orders in one model

Do this	What you should see
Open semantic model → Editing	D16-nakheel, with customers
Get data → Text/CSV → upload orders.csv	Next through the preview to Power Query
Save; apply changes if prompted	orders added to D16-nakheel
Return to the report → Refresh	Data lists customers and orders

For later cleaning: Open semantic model → Editing → Transform data. Finish with Save, then report Refresh.

EVERYONE DO

Step 7.2

Your turn, from the file to the card

Download customers.csv and orders.csv from shared/datasets/nakheel.
Follow activity 10, steps 1–5, with its browser boxes if needed.

power-query/labs/D16/10-Evr-PowerBI-End-To-End/README.md

STUDENT DO

One more chart

Completed revenue by channel. Then, if you have time, completed revenue by month.

power-query/labs/D16/10-Evr-PowerBI-End-To-End/README.md

STUDENT DO

Revenue by channel, completed orders only

INSTRUCTIONS	Clustered column chart: channel on the X-axis, Sum of order_total on the Y-axis. Add the filter status = completed. Check: the two bars add up to 316,023.50.
HINT	Which rows is this chart summing? The card’s filter does not apply here.
BONUS	Line chart by month (lab 11, step 5): which month was the best?

Full steps and Check yourself: power-query/labs/D16/10-Evr-PowerBI-End-To-End/README.md

REVIEW

Time's up. Let's review.

From a CSV file to a chart

Solved: power-query/labs/D16/10-Evr-PowerBI-End-To-End/Solved/solution.md

THE ANSWERS

Every number matches SQL

Visual	Filter	Value
Bar chart: customers by segment	none	Retail 435 · Wholesale 65
Card: Sum of order_total	status is completed	316,023.50
Columns: revenue by channel	status is completed	store 205,338.00 · online 110,685.50

Save D16-nakheel-first-report. Power Query Home → Advanced Editor shows Text.Trim and Text.Proper behind your clicks.

BigQuery ML functions today

In class	What it gives you	Activity
CREATE MODEL	trains and saves a model; model_type picks the kind	01–04
ML.EVALUATE	the scores; for ARIMA_PLUS, on the rows you pass	01–04
ML.PREDICT	a prediction for each row; inside a view, at every read	01–03
ML.FORECAST	the next days, each with a lower and upper bound	04
ML.TRAINING_INFO	one row per training round: loss and time	05

Lab	New in the labs
06	**ALTER MODEL (description, labels) · ML.PREDICT on two versions**
07	**ML.DESCRIBE_DATA · ML.VALIDATE_DATA_SKEW · ML.VALIDATE_DATA_DRIFT**
08	**ML.FORECAST on a retrained model · a view swapped to it**
09	**MATRIX_FACTORIZATION · ML.RECOMMEND (needs a reservation)**

EVERY MODULE 3 FUNCTION · resources/bigquery-ml-functions.md

CLOSE

What you can do now

- Pick the model; score it honestly
- Five steps, four models, four views
- Say what MLOps is and how BigQuery does it
- Know your licence; chart a CSV

NEXT · D17 More files: append, a folder, merge.

AFTER CLASS · SELF-GUIDED · NOT SUBMITTED

Labs 06 to 11

Lab	Folder	What you try
06	labs/D16/06-Lab-MLOps-Versions-And-Promotion	two diamond models, a model log, the better one promoted behind the view
07	labs/D16/07-Lab-MLOps-Monitoring-Drift	describe the data, find skew and drift, the retrain-if-needed script
08	labs/D16/08-Lab-MLOps-Retrain-And-Serve	retrain on all 15 years, forecast 2015, table or view
09	labs/D16/09-Lab-Matrix-Factorization-Nakheel	optional: recommend Nakheel products (needs a reservation)
11	power-query/labs/D16/11-Lab-Power-Query-First-Clean	clean city, a conditional column, revenue by month